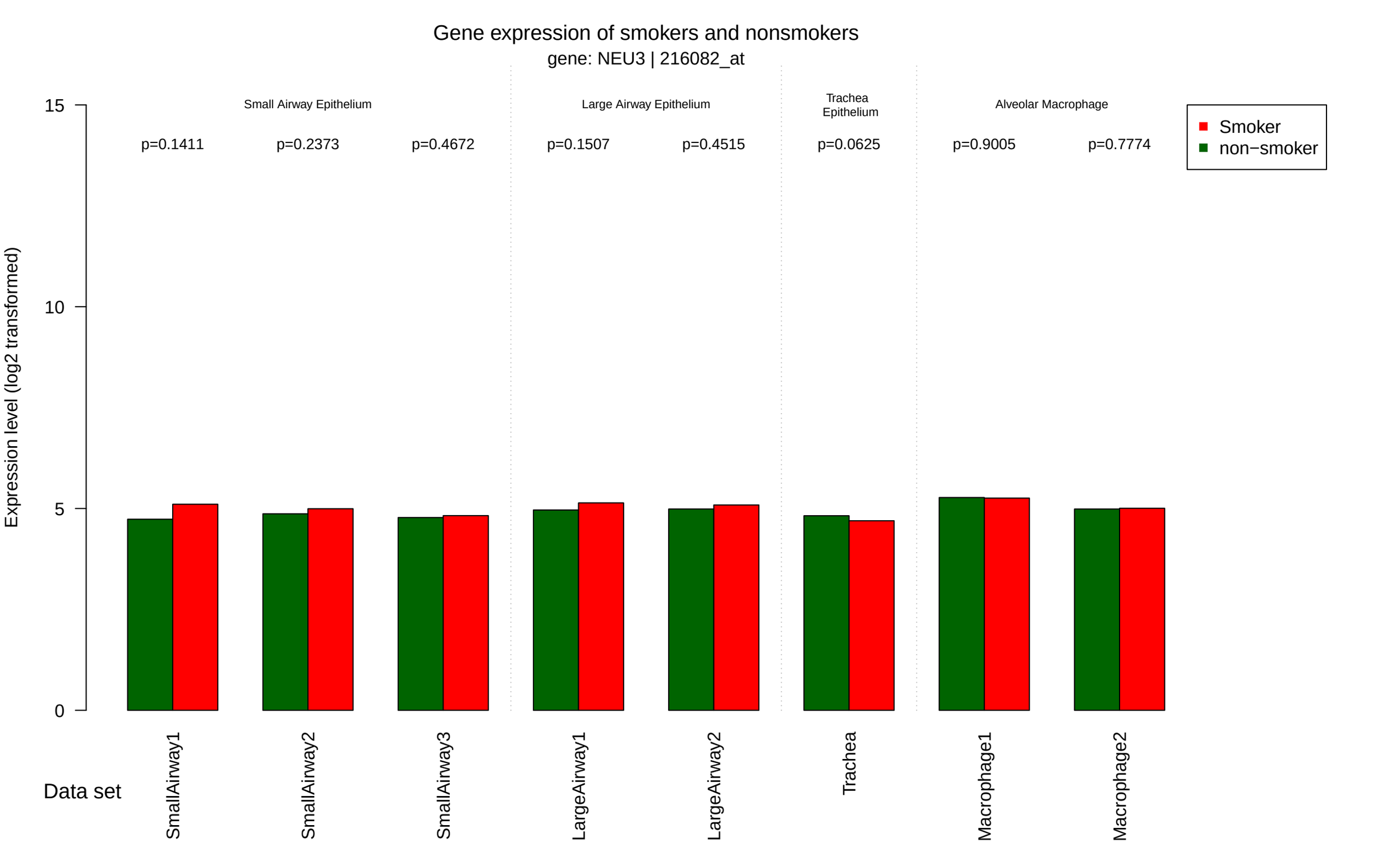
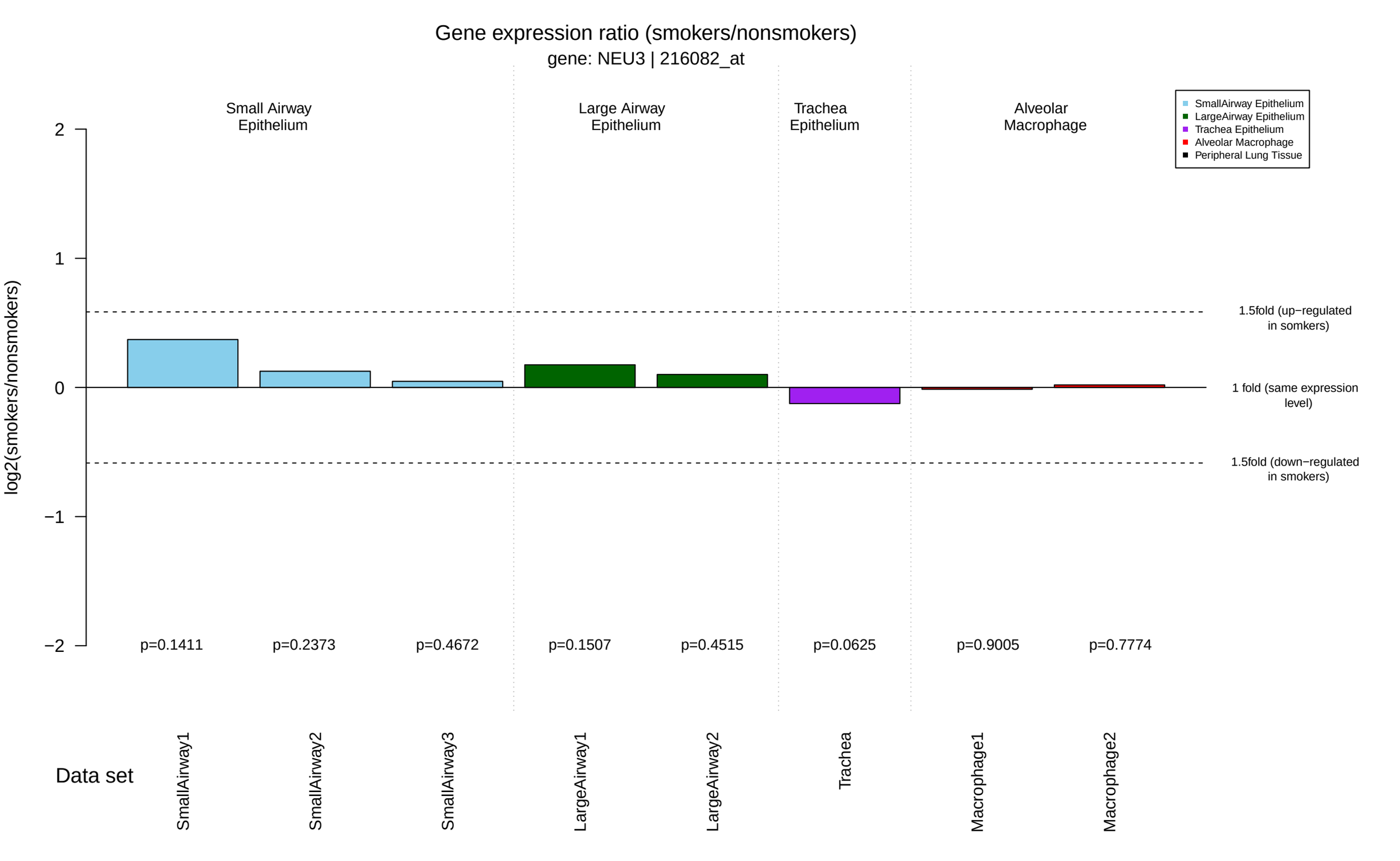
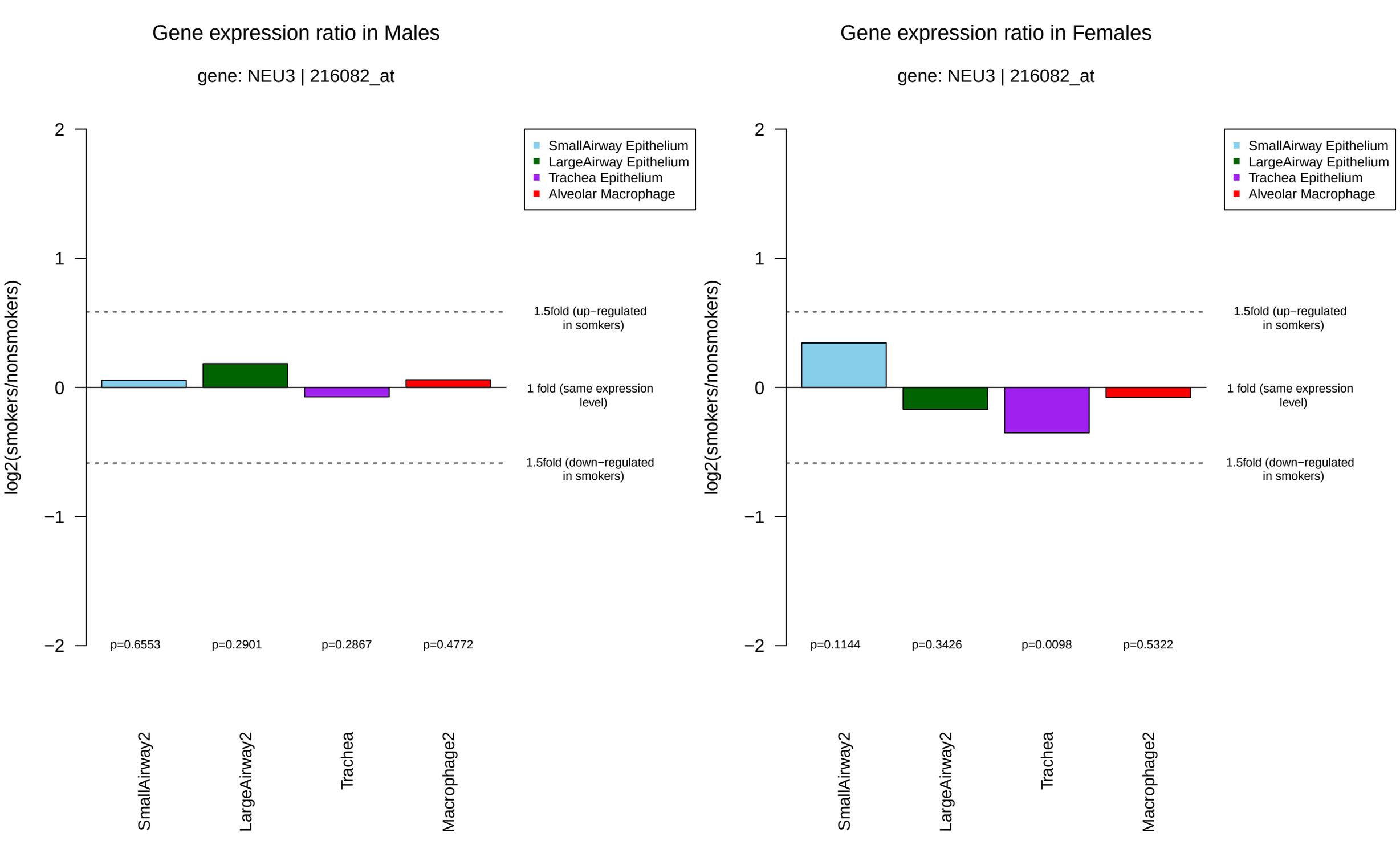
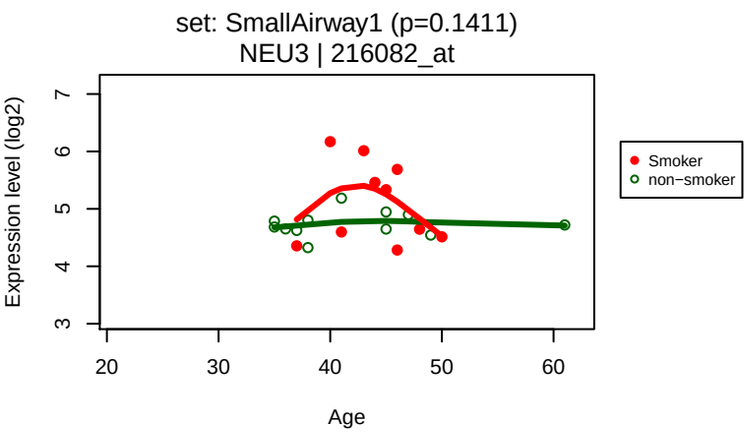
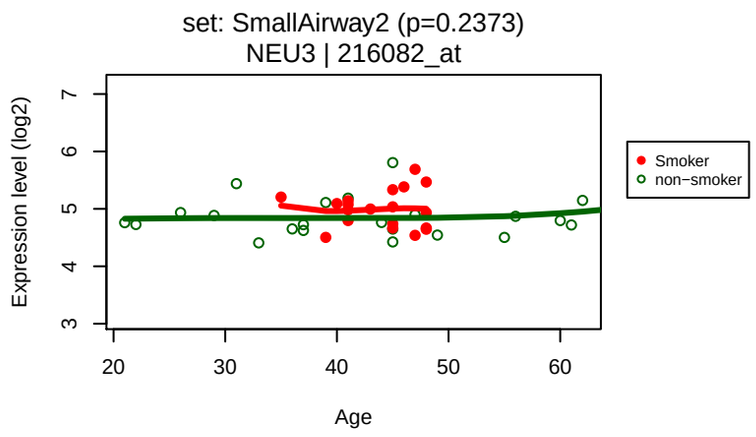
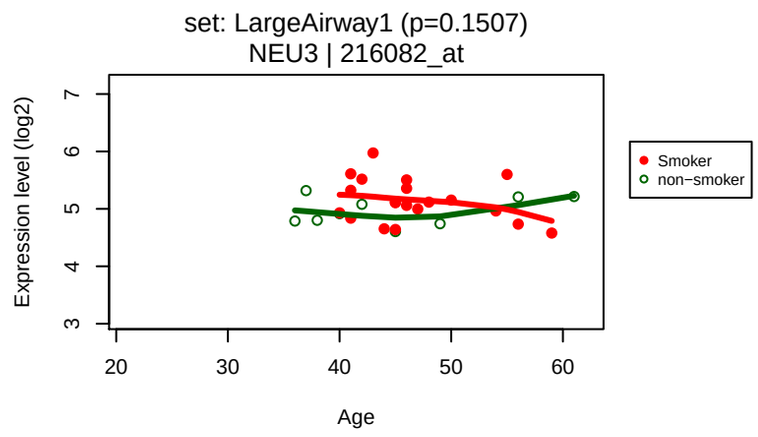

Gene expression of smokers and nonsmokers
gene: NEU3 | 216082_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1411
p=0.2373
p=0.4672
p=0.1507
p=0.4515
p=0.0625
p=0.9005
p=0.7774
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

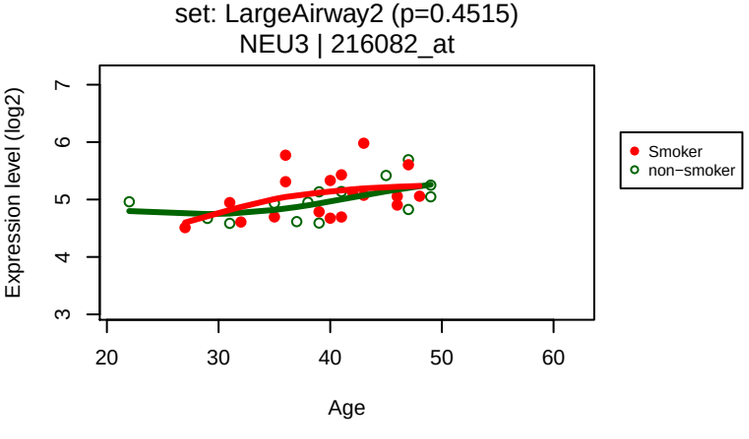

Gene expression ratio (smokers/nonsmokers)
gene: NEU3 | 216082_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1411
p=0.2373
p=0.4672
p=0.1507
p=0.4515
p=0.0625
p=0.9005
p=0.7774
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

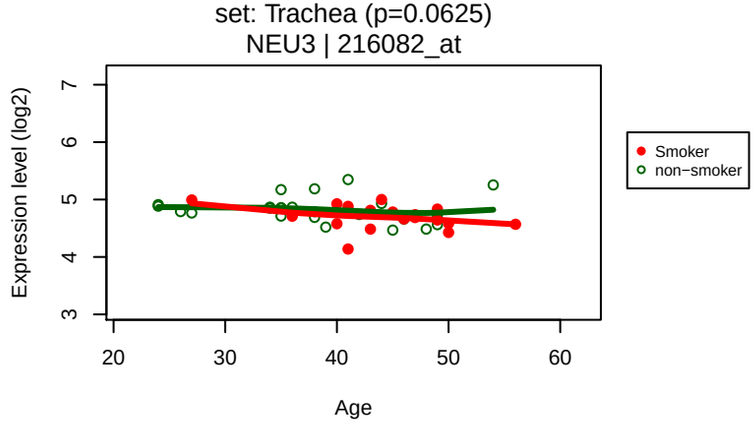

Gene expression ratio in Males
gene: NEU3 | 216082_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6553
p=0.2901
p=0.2867
p=0.4772
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEU3 | 216082_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1144
p=0.3426
p=0.0098
p=0.5322
SmallAirway2
LargeAirway2
Trachea
Macrophage2

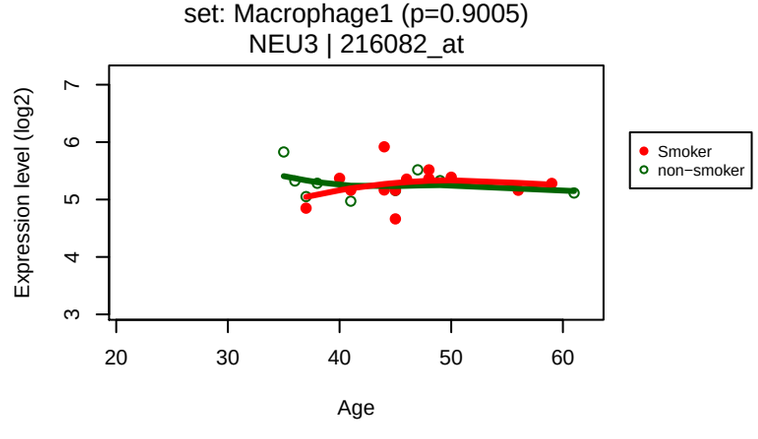

set: SmallAirway1 (p=0.1411)
NEU3 | 216082_at
Expression level (log2)
Age
Smoker
non−smoker

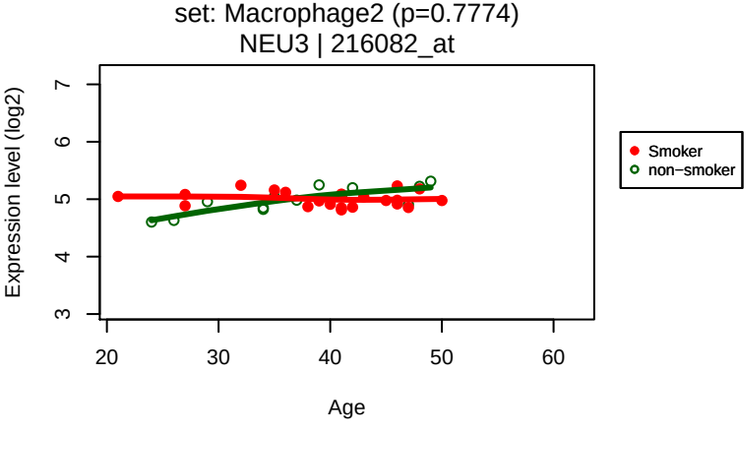

set: SmallAirway2 (p=0.2373)
NEU3 | 216082_at
Expression level (log2)
Age
Smoker
non−smoker

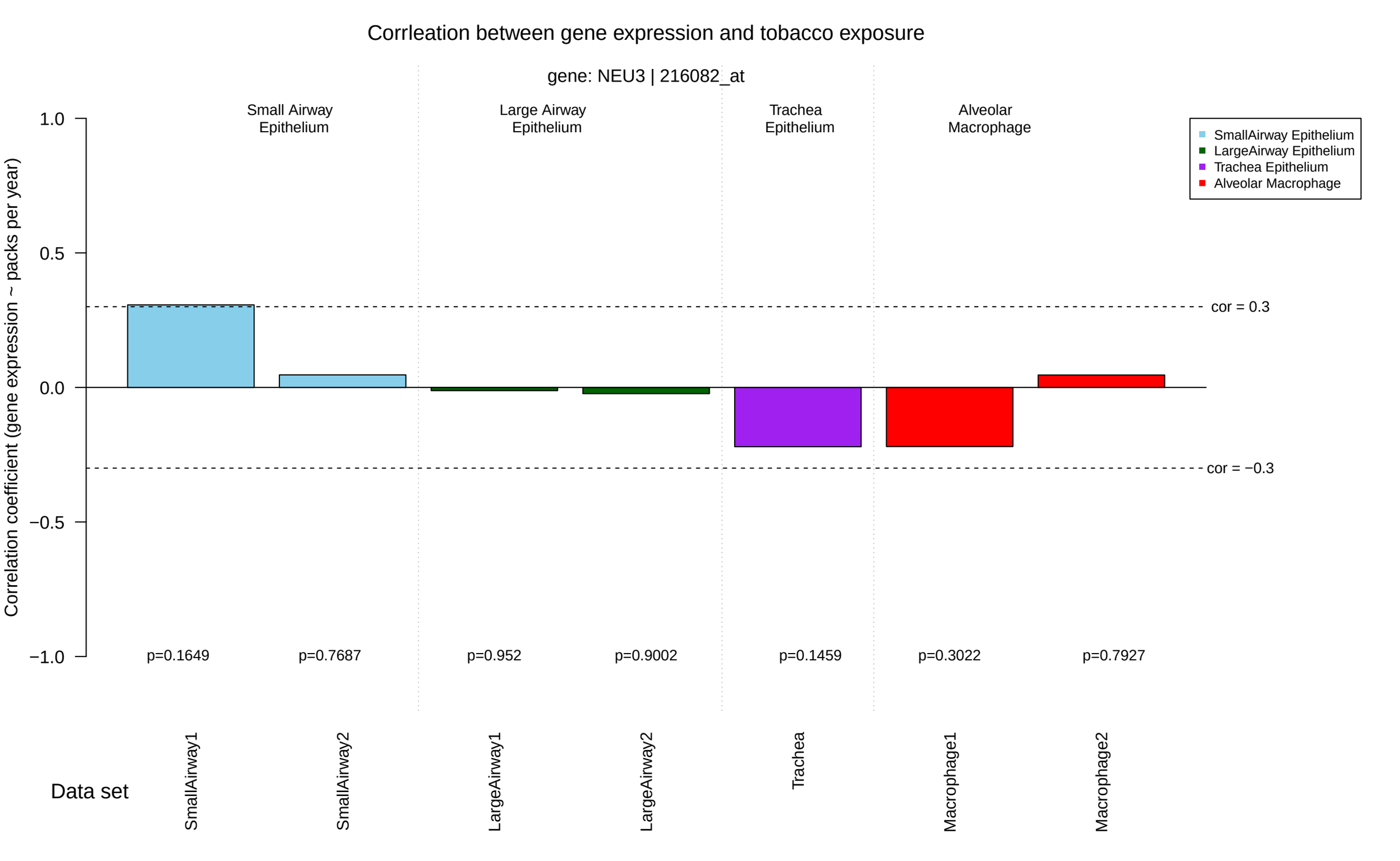

set: LargeAirway1 (p=0.1507)
NEU3 | 216082_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4515)
NEU3 | 216082_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0625)
NEU3 | 216082_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9005)
NEU3 | 216082_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7774)
NEU3 | 216082_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEU3 | 216082_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1649
p=0.7687
p=0.952
p=0.9002
p=0.1459
p=0.3022
p=0.7927
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2